Do parkland trees help or hurt the crops beneath them?

A trait-based ecohydrological model for semi-arid agroforestry

Cella Schnabel · Larsen Lab · February 24, 2026

Laikipia, Kenya — stochastic soil-moisture framework (Rodriguez-Iturbe & Porporato 2004)

The puzzle

Field evidence is contradictory:

- *Faidherbia albida* (winter-thorn acacia) → consistently facilitates maize in the Sahel
- Most evergreen parkland trees → mixed results; competition documented under drought
- Reviews report +5–30% yield gains — but also yield *losses*

Why?

> Trees interact with crops through multiple simultaneous mechanisms.
> Whether the net effect is positive or negative depends on
> **which traits the tree has** and **what the rainfall looks like.**

What the field evidence shows — and doesn't explain

The average picture is strongly positive

- Sub-Saharan meta-analysis (126 studies, 1,106 obs.): **crop yield ~2× higher** in agroforestry vs. monoculture *(Agronomy Sustain. Dev. 2019)*
- *Faidherbia albida* specifically: **2–4× yield** of unfertilised plots across the Sahel *(Garrity et al. 2010)*
- Agroforestry facilitative in **68–77% of study cases** in East Africa *(Kuyah et al. 2023)*

Hydraulic redistribution is documented

- Deep-rooted parkland trees lift **35–47 mm yr^{-1}** (8–10% of MAP) in Sahelian parklands *(Bargués-Tobella et al. 2014, WRR)*
- Tracer water applied to shrub roots reaches intercropped millet within **12–96 hours** *(Frontiers Env. Sci. 2018)*

But the field literature cannot explain *why* outcomes vary

Study system

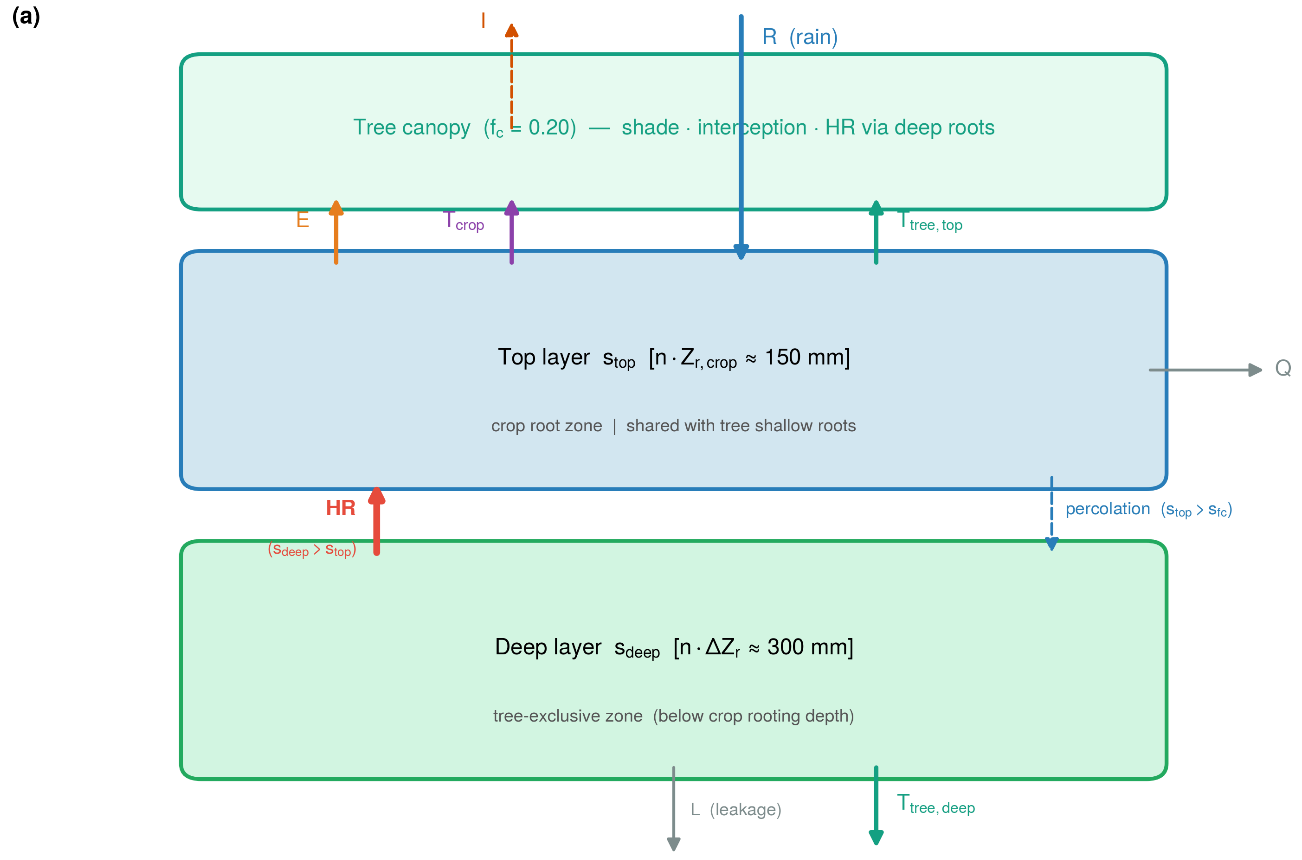
Location: smallholder maize farms, Laikipia Plateau, Kenya (semi-arid, ~500–700 mm/yr)

Published model baseline: Krell & Caylor (UCSB) — stochastic ecohydrology model, validated against Kenya Seed Co. yield data

Extension: add a tree component and ask — for three contrasting tree archetypes across three rainfall regimes, when does the tree help vs. hurt?

Archetype	Phenology	Root depth	Canopy cover
Faidherbia albida	**Reverse** — leafless during crop season	1500 mm	20%
Evergreen deep	Evergreen	1500 mm	20%
Evergreen shallow	Evergreen	600 mm	30%

The two-layer model



Four tree–crop interaction mechanisms

Competition

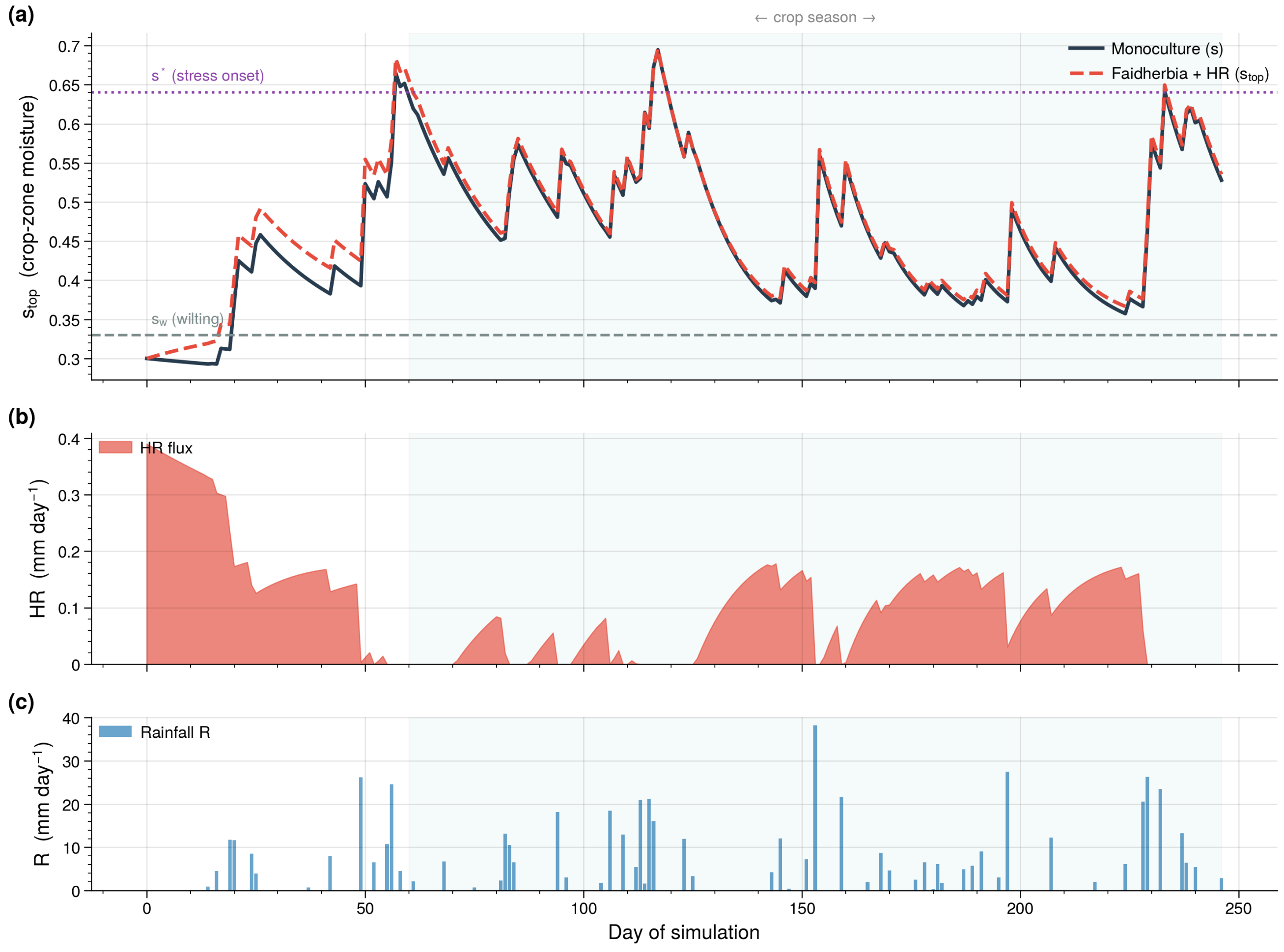
- **Shallow water competition** — tree roots draw from the same crop-zone bucket

Facilitation

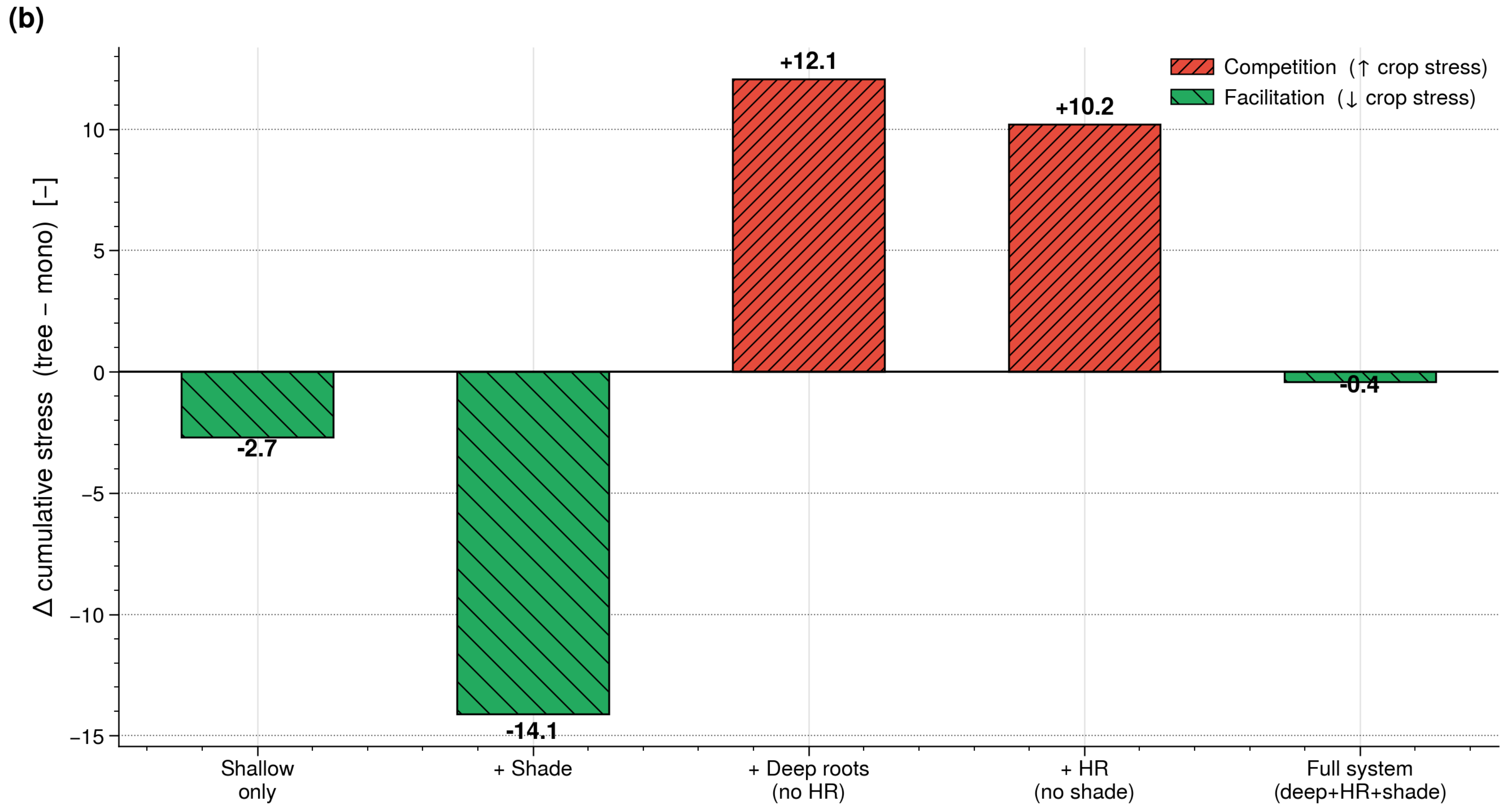
- **Shade** — tree canopy reduces soil evaporation (*E*) and transpiration via lower VPD
- **Deep roots** — tree preferentially fills demand from below the crop zone (deep-first allocation), leaving crop-zone water for the crop
- **Hydraulic redistribution (HR)** — at night, roots passively lift water from the deep layer into the crop zone along a water-potential gradient

> *Faidherbia adds a fifth lever:* ***reverse phenology*** *— no leaves during the crop season means zero competition and zero shade during the period that matters most.*

A single season: Faidherbia in action

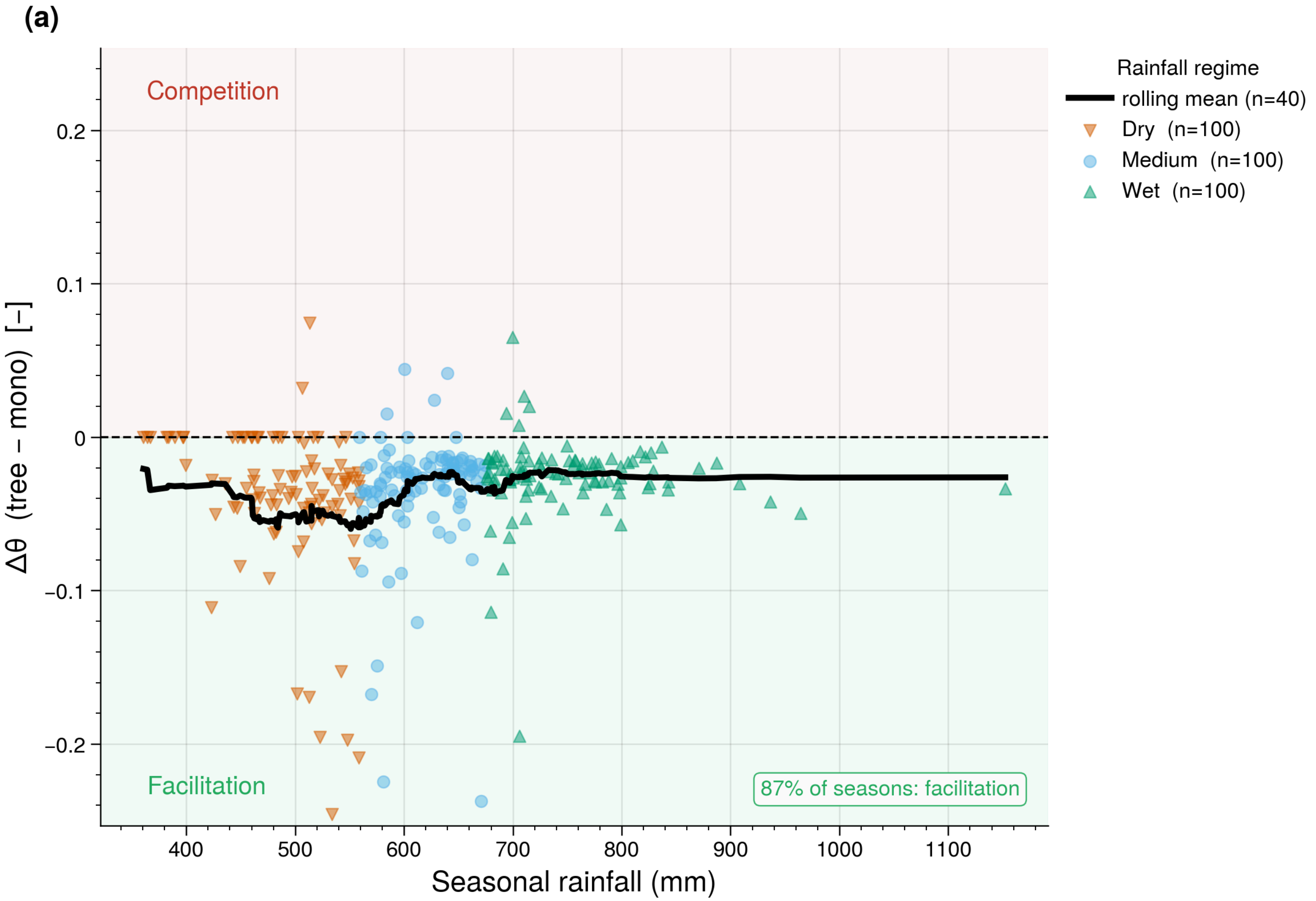


Which mechanisms drive facilitation?

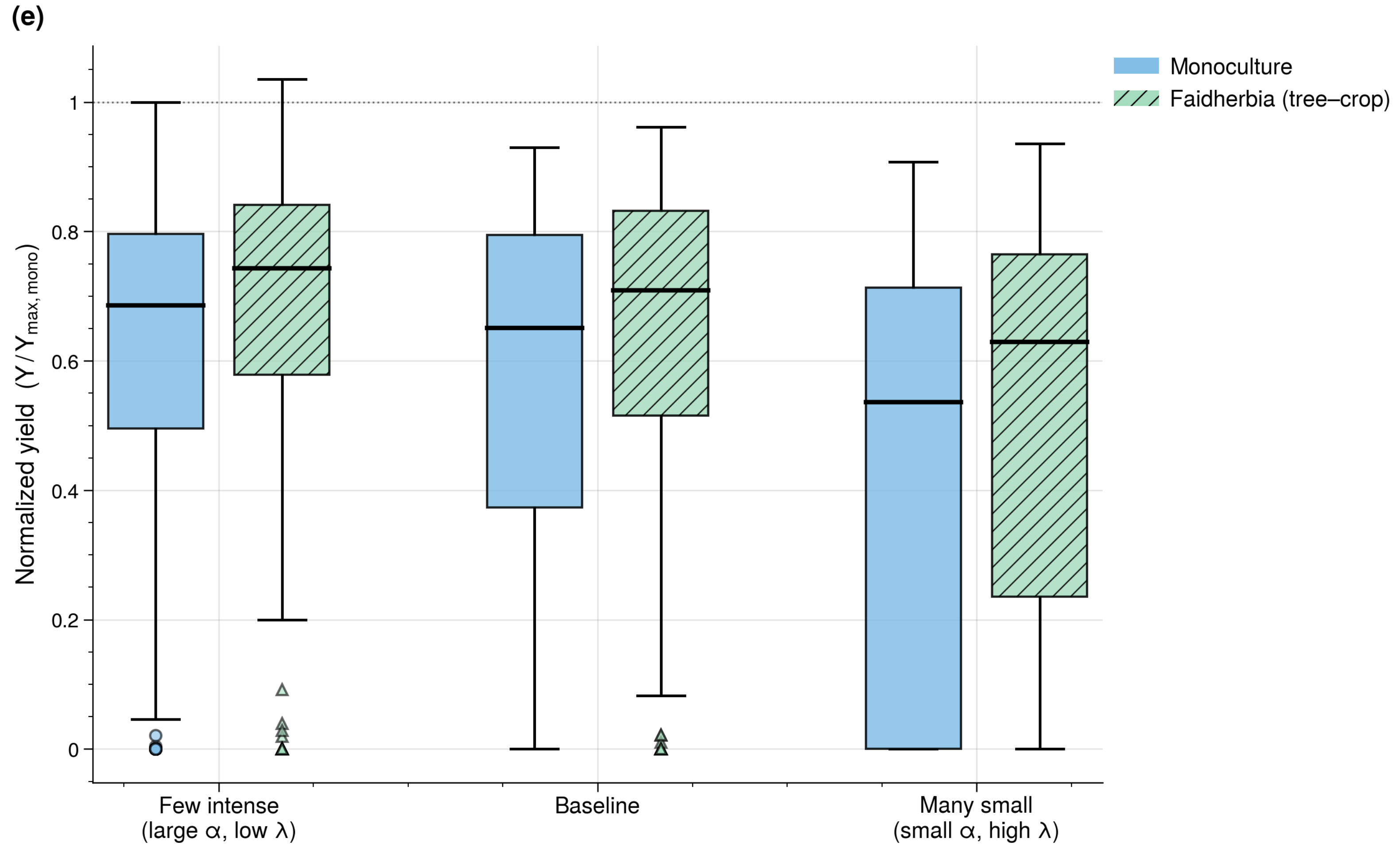


Factorial run: single season, evergreen tree (kc=1.0, cc=0.2, Zr=1500 mm, T_MAX=5.0 mm/day)

Faidherbia facilitates in 87% of seasons

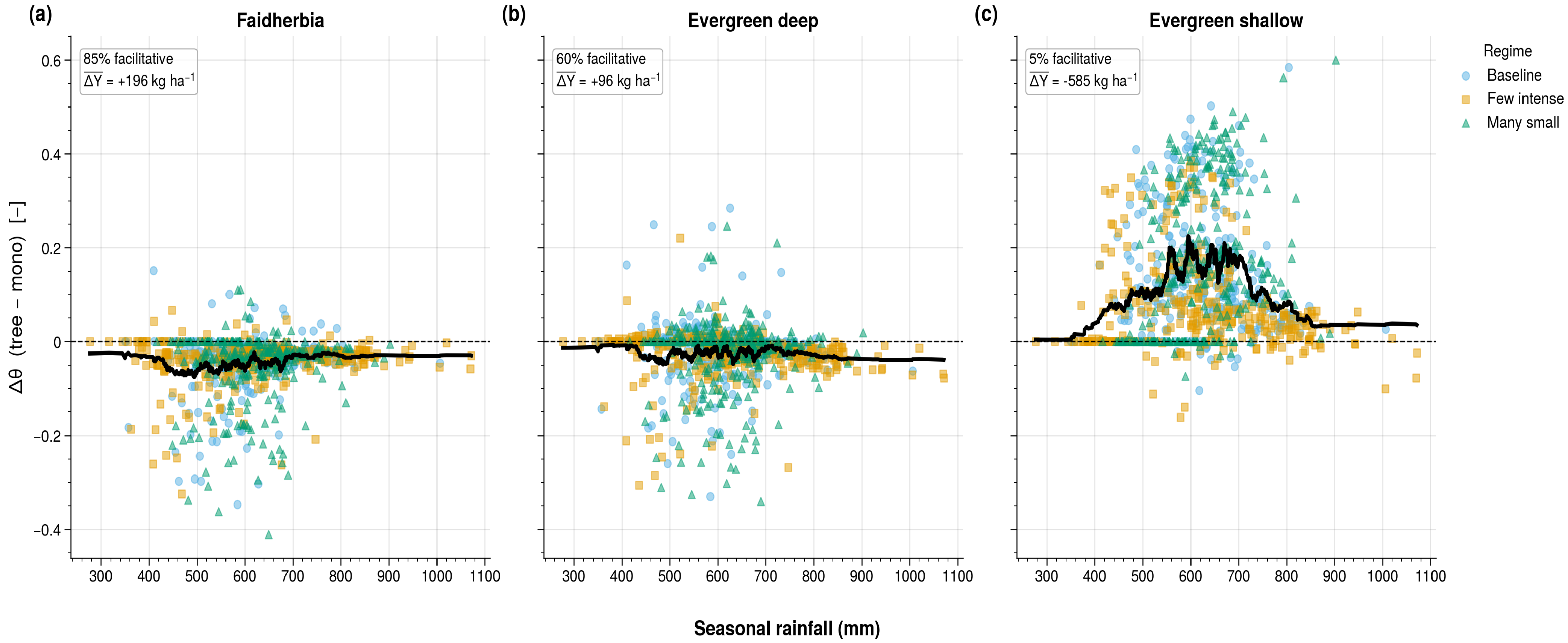


Yield gains hold across all rainfall regimes



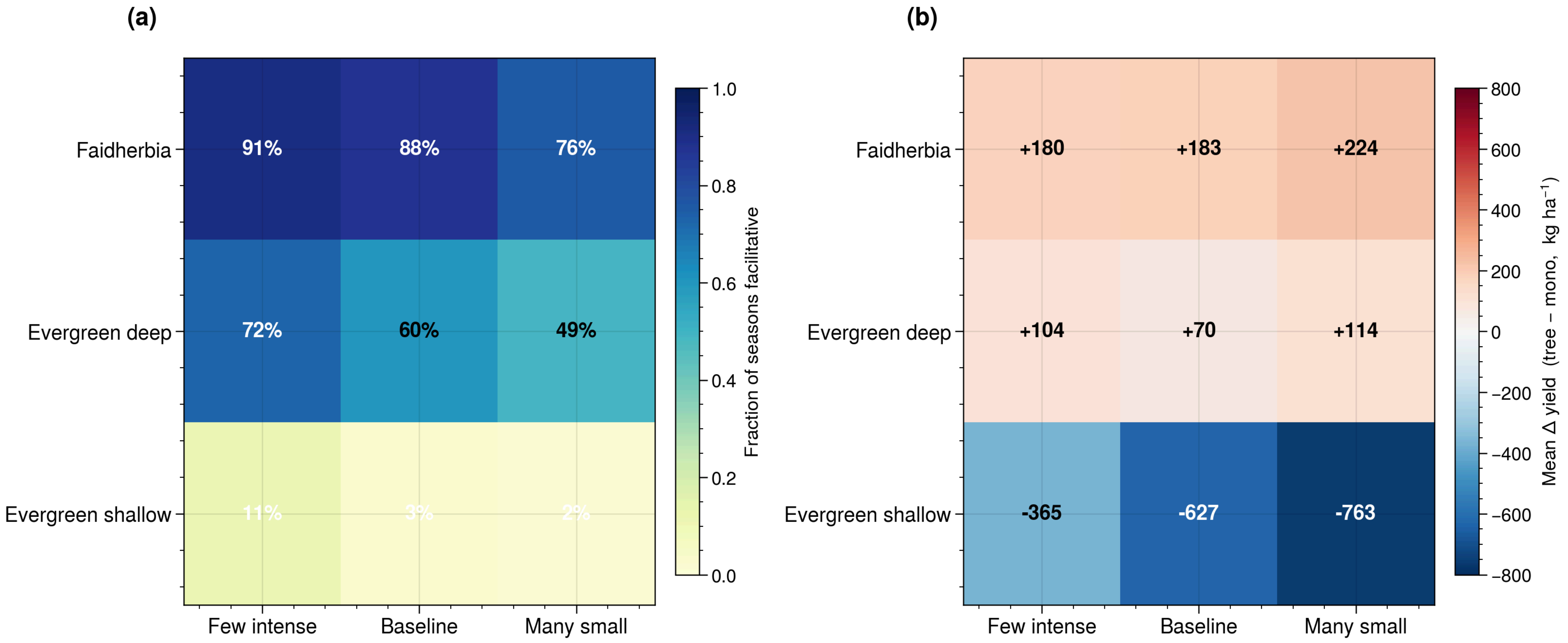
200 simulated seasons × 3 rainfall regimes · Faidherbia albida

Trait space determines the outcome



300 seasons × 3 rainfall regimes per archetype (n=900 season-pairs per archetype)

Summary: archetype × regime matrix



300 seasons × 3 rainfall regimes per archetype

Key findings

1. **Phenology is the master switch.** *Faidherbia*'s reverse phenology eliminates competition and shade during the crop season — facilitative in **85–91%** of seasons across all rainfall regimes (+180 to +224 kg ha^{-1}).
2. **Deep roots help — but only with HR.** Deep roots alone intensify competition (tree has more water, draws more). HR is what converts deep-root access into a crop benefit.
3. **Shade is net facilitative via E reduction**, but only when water is the primary limiting factor (semi-arid conditions). Shade and deep roots reinforce each other.
4. **Shallow-rooted evergreen trees are reliably competitive** (–365 to –763 kg ha^{-1}; only 3–11% of seasons facilitative). Shallow roots + year-round demand = direct competition.
5. **Rainfall regime modulates magnitude, not direction** — for Faidherbia and evergreen-deep, facilitation holds regardless of regime. For evergreen-shallow, competition holds regardless.

Next steps

Immediate

- **Sensitivity sweep:** vary `hr_max` (0.1–2.0 mm/day) and `canopy_cover` (0.1–0.4) — where does the competition→facilitation boundary sit in parameter space?

Short-term

- **Nitrogen fixation:** add `Y_MAX × (1 + N_bonus)` term — *Faidherbia* fixed-N is estimated at +5–15% yield gain in the field; model predicts ~10% from HR alone, so N-fixation could double the benefit
- **Validation:** compare Δ yield predictions to Bayala et al. and Garrity et al. field data

Longer-term

- **Deciduous vs. evergreen phenology sweep** — continuous trait rather than discrete archetypes

Thank you

Questions?

Figures and model code: `tree.ipynb`

Framework: Rodriguez-Iturbe & Porporato (2004), Laio et al. (2001), Porporato et al. (2001)

Agroforestry references: Bayala & Prieto (2020), Bargués-Tobella et al. (2014), Garrity et al. (2010)